

KRUSHNA SUTARIYA

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ABOUT ME

Software Engineer with hands-on experience in React, TypeScript, Python, FastAPI, Docker, CI/CD, Git, and cloud-based application development. At BMW Group, built real-time visualization tools for ADAS and localization data using React, TypeScript, Python, WebSockets, Chart.js, Plotly, and Foxglove Studio. Developed and deployed a full-stack portfolio platform with a React frontend, FastAPI backend, WebSocket-based AI assistant, Docker deployment, and GitHub Actions automation. Recently submitted an M.Sc. thesis on explainable LLM-based recommendation systems and completed cloud and DevOps projects using AWS, GCP, Terraform, and CI/CD pipelines.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java basics

Frontend: React.js, Vite, HTML5, CSS3, Tailwind CSS, Mapbox GL JS, React Leaflet, Chart.js, Turf.js, UI/UX

Backend and APIs: Python, FastAPI, REST APIs, WebSockets, Pydantic, Resend API, SQL Server, DynamoDB

Cloud and DevOps: AWS, GCP Cloud Run, Docker, Docker Compose, Terraform, GitHub Actions, GitHub Pages, GitLab CI/CD, Git, Linux, pytest

AI and Machine Learning: LLM concepts, Gemini API, Explainable AI, recommender systems, Scikit-learn, Pandas, NumPy

Tools: Foxglove Studio, Figma, Jira, Postman, Google Colab, VS Code

PROFESSIONAL EXPERIENCE

BMW Group

Software Development Intern

Feb 2025 – Jul 2025

Munich, Germany

- Redesigned the web-based Localization and Map Visualization platform for BMW's ADAS team using React.js, TypeScript, and Mapbox GL JS, including modular UI components, collapsible sidebars, layer controls, and dynamic map-state handling, resulting in a 40% increase in data analysis speed for test-drive validation teams.
- Built dual-layer geospatial visualizations for lane and road position candidates using Mapbox GL JS and Turf.js, including color-coded paths and timestamp hover popups.
- Developed a Foxglove Studio configuration panel with React.js and TypeScript for live layer visibility control of traffic signs, lane markings, centerlines, and map settings.
- Replaced static PDF optimization plots with a real-time browser dashboard using Python, WebSockets, Plotly, Matplotlib, and Chart.js frontend visualizations.
- Implemented 2D and 3D vehicle localization hypothesis visualizations, including WGS84-to-ENU coordinate conversion and probability-scaled cylinders in Foxglove Studio.
- Worked in an agile Scrum environment with sprint planning, daily stand-ups, retrospectives, Git version control, and GitHub pull request reviews.

Techsphere Softwares LLP

Front-End Developer

Jan 2021 – Apr 2023

India

- Developed responsive web pages and reusable UI components using HTML, CSS, and JavaScript for client-facing applications.
- Integrated frontend views with backend APIs and supported usability improvements, browser compatibility testing, and clean code maintenance.
- Collaborated with developers and stakeholders to translate requirements into functional web interfaces.

Savaj Infotech

Web Designer

Aug 2016 – Dec 2020

India

- Designed and implemented website layouts, landing pages, and UI components with focus on usability, responsiveness, and visual consistency.
- Created UI concepts and page structures for client projects and collaborated with developers to deliver functional web solutions.

PROJECTS AND RESEARCH

Full-Stack Personal Portfolio Website

Jun 2026

React.js, Vite, Tailwind CSS, FastAPI, WebSockets, Gemini API, Resend, Docker, GitHub Actions, GitHub Pages, Hugo, Face Spaces

- Built and deployed a full-stack personal portfolio with responsive React frontend, project showcase, downloadable CV, contact form, and a WebSocket-based Gemini AI assistant.

- Implemented real-time communication using WebSockets and integrated AI-powered chat functionality using Gemini API and structured project context.
- Developed a FastAPI backend with REST endpoints, Pydantic validation, Resend email notifications, Docker Compose setup, and deployment using GitHub Pages for the frontend and Hugging Face Spaces for the backend. Live demo: krushnasutariya.github.io/krushna-portfolio

SkyMap Weather Application
React.js, Vite, Tailwind CSS, React Leaflet, OpenWeather API, GitHub Pages

Jun 2026

- Built and deployed a responsive weather map application with city search, interactive map selection, live weather data, and custom weather UI components.
- Integrated external APIs and map-based interactions to display weather, forecast details, location-based data, and user-friendly visual feedback. Live demo: krushnasutariya.github.io/sky-map-weather

Master’s Thesis: Explainable LLM-Based Recommendation System
Hochschule Heilbronn, Germany

Nov 2025 – Apr 2026

- Designed explainable recommendation strategies for the IdeaLize academic collaboration platform using textual, keyword-based, and visual explanation formats.
- Compared textual, keyword-based, and visual explanation formats to improve transparency and user trust.
- Built a high-fidelity Figma prototype and conducted a user-study-based evaluation to analyze trust, clarity, usefulness, and user perception of explanations.
- Applied Explainable AI principles, recommender-system concepts, and human-centered interface design.

DevOps and SecOps CI/CD Automation Project
Python, Docker, GitHub Actions, pytest, Google Cloud Run

Academic Project

- Containerised a Python REST API and configured automated CI/CD workflows with GitHub Actions, pytest, and code coverage checks.
- Deployed the application to Google Cloud Run using an automated image build and cloud deployment workflow.

Serverless Online Grocery Application
AWS, Terraform, Lambda, S3, DynamoDB, API Gateway, GitLab CI/CD

Academic Project

- Built a serverless web application architecture on AWS using Terraform for infrastructure provisioning and automated deployment.
- Implemented IAM, S3, DynamoDB, Lambda, and API Gateway, supported by a GitLab CI/CD pipeline.

Movie Recommendation System
Python, Pandas, NumPy, Scikit-learn, Google Colab

Academic Project

- Built a recommendation system using collaborative and content-based filtering concepts to generate personalized movie suggestions.
- Processed datasets and evaluated recommendation logic using data preprocessing and similarity-based matching.

EDUCATION

Hochschule Heilbronn <i>Master of Science in Software Engineering</i> Current average grade: 1.7; thesis submitted Apr 2026	Mar 2024 – Expected 2026 Germany
Shree Swami Atmanand Saraswati Institute of Technology (SSASIT) <i>Bachelor of Engineering in Computer Engineering</i> CGPA: 8.36 / 10	Jun 2013 – May 2016 India
Government Polytechnic for Girls <i>Diploma in Computer Engineering</i> CGPA: 8.64 / 10	Jun 2010 – May 2013 India

LANGUAGES

English - Fluent C1 | German - A2 / B1 intensive course in progress | Gujarati - Native | Hindi - Native

CERTIFICATIONS

AWS IoT | DevOps on AWS | Fundamentals of Machine Learning and AI | Intermediate Machine Learning